

Security and Trust in Cryptocurrencies

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Abstract

The term of cryptocurrency is phenomenon of recent years, standing for digital currency based on principles of cryptography. General public may confuse this term with Bitcoin, but many other cryptocurrencies arose. Basic principles of all cryptocurrencies are similar and are briefly introduced in this paper. Low fees, possibility to make virtually anonymous payments without involving banks, and expected high level of protection of personal data are benefits associated with holding using cryptocurrencies. High volatility, cryptocurrency e-wallet thefts and possible anonymous criminal activities funding are mentioned as the main disadvantages of cryptocurrencies. Issues of security of cryptocurrencies mining, holding and transferring are investigated. Furthermore, basic arrangements for building trust in cryptocurrencies are discussed.

Key words: cryptocurrency, virtual currency, trust, Bitcoin, Litecoin

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1 Introduction

The term of cryptocurrencies exist in praxis only since year 2009 when the first cryptocurrency was introduced named Bitcoin by Satoshi Nakamoto (pseudonym). Cryptocurrencies represent a digital medium of exchange allowing secure, decentralized and distributed economic transactions. Cryptocurrencies integrate principles of cryptography with electronic money. Sometimes they are referred as virtual currencies, however virtual currency does not have to use principles of cryptography in general. The basic principle of cryptocurrencies is that no one may accelerate or significantly abuse their production. Certain predefined amount of cryptocurrency is produced collectively by the entire cryptocurrency distributed system and therefore rate of production is determined by a value defined in prior and publicly known. Cryptocurrency allows virtually costless transfers of coins (cryptocurrency units) between clients in a computer peer-to-peer network.

Bitcoin was first cryptocurrency introduced and still it is the most popular. All other cryptocurrencies are inferred on principle from Bitcoin with some adjustments. There are two types of Bitcoin users: basic ordinary users and so called “miners”. Ordinary Bitcoin users use digital wallet similar to electronic banking for management of Bitcoin cash and transferring

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payments in bitcoins. Bitcoins exist only as data in a computer or mobile device. Access to these data has only the holder of cryptographic private key. If the wallet data is lost (e.g. the wallet data file is inadvertently corrupted or deleted), then the bitcoins contained in this wallet are lost forever (when file was not backed up). The public address of the wallet still exists, but it can only be accessed by the private key, which has been deleted. Unless someone breaks highly secure encryption built into the cryptocurrency system, then it is not possible to recover the lost bitcoins. And breaking Bitcoin's encryption is virtually impossible using common computational force of personal computer in timely manner.

New units of Bitcoin are generated by Bitcoin network in process called mining, which is performed by so called miners. Miners are dedicating their computational power to solving artificial mathematical problem. Transactions are packed in a block that fits very strict cryptographic rules that will be verified by the network. Bitcoin uses SHA2-256 cryptographic algorithm as proof of work mechanism during transactions confirmation. These cryptographic rules prevent previous blocks from being modified because doing so would invalidate all following blocks. No individual can control what is included in the block chain or roll back own transactions.

Many other cryptocurrencies emerged since introduction of Bitcoin, but the most are derived from specifications of the Bitcoin. Some cryptocurrencies (e.g. Peercoin) include also mechanisms that allow fractions of cryptocurrency units being assigned to their holders as form reward for holding the cryptocurrency, what can be considered as analogy of interest. The most cryptocurrencies generate new units of currency up to a preset maximum amount of currency that will ever be generated. This should insure the scarcity and prevent high inflation.

Cryptocurrencies provide almost perfect anonymity of transactions, what allows avoidance of authorities' attention. This fact attracts various users to cryptocurrencies, for example several drug dealing electronic marketplaces were accepting payments in cryptocurrencies. Another issue of cryptocurrencies is their high volatility and potential for price manipulations. Some cryptocurrencies were unfairly manipulated by pre-mining, so currency was generated by its creators before mining code is released to the general public.

Interchangeability of cryptocurrencies for official currencies is in limited extent, basically only Bitcoin and Litecoin are directly changeable at virtual markets (e.g., Vircorex, BTC-e, BTC-e etc.). Other cryptocurrencies can only be traded for Bitcoin or Litecoin and then for official currencies. Banks generally do not offer services attached to cryptocurrencies and sometimes also refuse to provide services to cryptocurrency companies.

2 Literature review

The theoretical and empirical research of cryptocurrencies is no of the big extent, because this topic is recent and novel. Bitcoin as the most spread representative of cryptocurrencies is the main concern of authorities trying to regulate alternatives to official currencies. Chinese central bank forbid to financial institutions in country to offer services related to Bitcoin. Internal Revenue Service, the government tax administration authority in USA, treats virtual currencies as property (not currency) for federal tax purposes and its amount is therefore taxable (IRS, 2014). In other countries virtual currencies have also problems with acceptance. Authorities in Norway do not recognize Bitcoin (and other cryptocurrencies) as currency while it does not fulfill requirements of being currency. Buying cryptocurrencies is in Finland considered as purchase of goods. Regulators in several other countries warn against use of virtual currencies, because legal

ways of acquiring the paid amount back, when some problem with object of transaction arises, do not exist (Matonis, 2013).

Krugman (2013) stated that Bitcoin is very good medium of exchange, but it does not store the value at reasonably stable level. Gold preserves value, while its value is limited by technology of its mining and its limited supply in mining sites. Federal Reserve Bank (Fed) guarantees the value of US dollar (in some limits – inflation etc.). On the case of cryptocurrencies, even though they are mainly limited in preset amount of units ever be created, but nobody guarantees their minimal value as long as they are decentralized and fully independent.

Chaum (1983) laid the theoretical foundations of cryptocurrencies, when he drafted use of cryptology for allowance of untraceable electronic payments. Nakamoto (2009) proposed the creation and then introduced Bitcoin network in early 2009. The acceptance of cryptocurrencies was investigated by Luther (2013) with conclusion that cryptocurrencies are unlikely to be widely accepted in the conditions of high volatility and absence of a government support. Morris (2014) adduced that newly introduced cryptocurrency might be manipulated by pre-mining or hidden launches for pre-selected users with high mining rewards, what might be a part of a cryptocurrency's original design.

Tucker (2013) remarked a high volatility of cryptocurrencies making the holding of cryptocurrencies very risky, while they might manipulated by artificially inflating the price of cryptocurrency through false positive statements, in order to sell the purchased cryptocurrency at a higher price (pump and dump scheme). Once the operators of the scheme sell out the cryptocurrency, the price falls and other holders of given cryptocurrency lose money. Milnes (2013) showed that the volatility of cryptocurrencies causes so called a deflationary bias. More and more people trying to acquire to a limited number of cryptocurrency increases its value. Then deflation encourages currency hoarding rather than spending or usage. For example O'Brien (2013) at this point suggests that cryptocurrency ceases to be a currency and becomes a virtual commodity.

Miller et al. (2014) adduced that the processing power used for cryptocurrency mining is wasted, rather than used for meaningful purpose. They proposed to use these computing resources more useful by archiving the meaningful data from the huge data fund, for example, backing up data from the Library of Congress (approximately 200 TB of data) in network of new virtual currency named Permacoin. The data would be stored locally, controlled by a network and the user would obtain corresponding permacoins for storing them. It would be a public archive of valuable data that could be useful for mankind. Investments in hardware and storage would actually serve to secure and distribute data in a decentralized system widely available with multiple data backups.

Hayes (2015) investigated factors of cryptocurrency value and found that the most of its value can be explained by the three variables: computational power (indirectly difficulty), coins per minute and which algorithm is used. Moore and Christin (2013) investigated security of cryptocurrency accounts. Their results indicated a high risk of breaching security of popular cryptocurrency exchanges. Later crash of Mt.Gox marketplace in 2014 confirmed their conclusions.

3 Factors of Trust in Cryptocurrencies

In this paper we have identified multiple factors that affect the trust of potential users in cryptocurrency.

Deem (2015) stated that any currency system seeking successful adoption must be in want of trust. Trust that a representation of value (such as a paper note) is backed by real value or

obligation to repay; trust that those representations will be accepted by others and trust that the representations of value are not counterfeit.

Early currencies contained precious metals, which people trusted to have an intrinsic value that they could use to obtain goods or services. Today, most countries use fiat currencies are not tied to any physical value such as precious metals. Instead, fiat currencies are backed by government guarantee and currency demands ultimate trust. The identity of those involved in a transaction (and the ability to verify such identity) is the basis of trust in currency used in transaction. In the case of fiat currency, the currency issuer and key players in a payment chain must be trustworthy to ensure that the currency is not counterfeit and that the currency will not be stolen before reaching its final destination. But in the case of cryptocurrencies, where value is established by algorithms and verified by the electronic transfer of data, all players in transactions are anonymous.

The trust in cryptocurrencies can be generated by the confidence in the technology which underlays the currency and the key market players have a high level of accountability. In fiat society generally trusts in the system because the banks and payment processors are regulated and enforced to retain a high level of security. Even if a security breach occurs, the parties involved will be held accountable and fiat currency users will get compensation.

However, in the case of cryptocurrencies, which are entirely decentralised, the value passes directly from payer to payee. The blockchain is extremely secure as it publically records every transaction that is ever made. Cryptocurrency system is secure in its technology, but it is marked many examples of hacking and fraud. Deem (2015) indicates that cryptocurrency trust issue lies with the gatekeepers between cryptocurrencies and their traditional fiat relations. Cryptocurrency-related businesses that act as interfaces with fiat currencies, such as exchanges and payment processors, are mostly unregulated. This is the spot where occur reported hacks, causing mistrust in the system and extreme volatility of cryptocurrencies. This mistrust comes out of the lack of accountability of these generally anonymous cryptocurrency-related businesses.

If fiat currency is stolen from a bank, its security is increased. Consequently, when a cryptocurrency exchange is compromised, the solution should be increasing the security of exchanges. The organisational and behavioral standards should be enforced by national regulators to increase security of the cryptocurrency industry and increase trust in cryptocurrencies.

3.1 Awareness

The awareness of cryptocurrency affects the numbers of its users and therefore also its stability. Significant role in the field of awareness play the media. High media coverage of Bitcoin's price rally definitely drew attention of numerous additional speculators. Besides the all forms of media also the word of mouth increases the awareness of cryptocurrencies and contributes to increase of cryptocurrency users' numbers.

On the other hand, negative information about cryptocurrencies in media discourages potential users, for example, the crash of Mt. Gox - former the biggest Bitcoin market (Hutchinson and Liew, 2014). It ceased its operations in February of 2014, because of big losses of Bitcoins and also official currencies. Faulty processes of market, which were known for longer period, but were not addressed by Mt. Gox management, caused multiple currency thefts. The fault subsisted in possibility to change identification of transaction on e-marketplace during its processing. Then impostor could claim that payment with original identification did not ascribe to his wallet. Mt. Gox customer care officials did not identify the payment with original identification and sent given amount again, even if such amount already has been sent there but it has only changed identification. This caused multiple losses and led to crash of Mt. Gox (Tanase, 2014). Its crash

significantly influenced also prices of Bitcoin at other e-markets and caused a disruption of trust in Bitcoin.

3.2 Legislation influence

Before year 2009 national legislations around the world did not consider virtual or cryptocurrencies. However rising amount of transactions in cryptocurrencies forced policymakers in many countries prepare legislation regarding the use of cryptocurrency in given economy. Broader allowance of cryptocurrency usage definitely supports its usage and builds trust in it, but the most authorities are trying to restrict or ban usage of cryptocurrencies as currency. Therefore usage of cryptocurrencies in legal economy is limited, but in illegal economy it keeps its status of safe and anonymous medium of exchange. Cryptocurrency legislation affects illegal transaction only in minor extent, while these transactions are already part of law-violating activity.

In general, the most legislative arrangements caused decreases in cryptocurrency exchange rates and led to negative effect on the trust in all cryptocurrencies. However regulation should be aimed to provide rules for cryptocurrency-based business to support responsible and accountable behavior to prevent frauds and hacks. In this case legislation would bring impulse for increasing trust in cryptocurrency area.

3.3 Availability

Availability represents the possibility of purchasing and interchangeability of cryptocurrency for official currencies. This is mainly enabled by private electronic markets available to general public for selling and purchasing various cryptocurrencies. Number of subjects accepting payments in given cryptocurrency also increases its availability or usability. Number of merchants accepting cryptocurrencies (mainly Bitcoin) is increasing, but they are facing the problem of high volatility of cryptocurrencies. Thus, they do not keep cryptocurrency for the longer period of time at most cases but change it for standard currency. Also the speed of transaction confirmation processes makes cryptocurrency more available and suitable for electronic payments. Furthermore, higher availability brings higher and more stable exchange rate of cryptocurrency, what raises trust in it.

3.4 Anonymity

In general, the cryptocurrency payments are anonymous and hardly traceable. This is the main reason of their usage in illegal activities. Anonymity and no central guaranty of currency are main reasons legislation restrictions of cryptocurrencies. Governments and other legal authorities are afraid of virtually untraceable transactions that might be connected with illegal activities or even financing terrorist organizations. Any cryptocurrency might be financial transfer to terrorists from their sponsors because both sides are anonymous. The lack of identification of transactions led to multiple legislation arrangements and restrictions of cryptocurrency usage in recent years. Authorities should also consider that even legal users might prefer anonymous transactions due to concerns of privacy intrusions by any third party not involved in transaction. Multiple cases of tracking of private communications make anonymity of cryptocurrencies very valuable and so anonymity of cryptocurrency affects supports the trust in cryptocurrencies.

3.5 Volatility

Volatility represents statistical measure of the dispersion of returns for a given security. Volatility can either be measured by using the standard deviation or variance between returns from that same security. In general, the higher the volatility means the riskier the security.

In order to investigate volatility of cryptocurrencies, two the most popular cryptocurrencies Bitcoin and Litecoin were selected, as long as they are direct interchangeable to standard currencies at public electronic markets.

Following figures show history of exchange rates of Bitcoin and Litecoin during last two years on major e-marketplace (BTC-e, 2015).

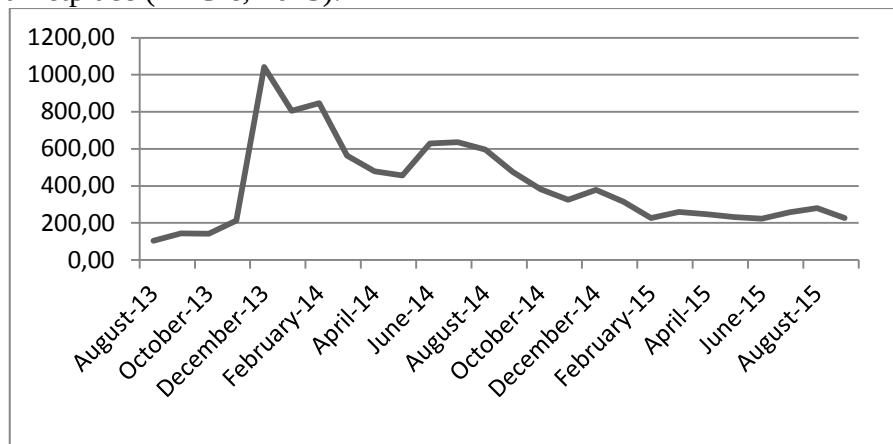


Figure 1 Bitcoin exchange rate at BTC-e marketplace

Figure 1 shows Bitcoin historical prices at BTC-e electronic marketplace in period since August 2013 until August 2015. High fluctuations of price indicate high volatility rate of Bitcoin in given period.

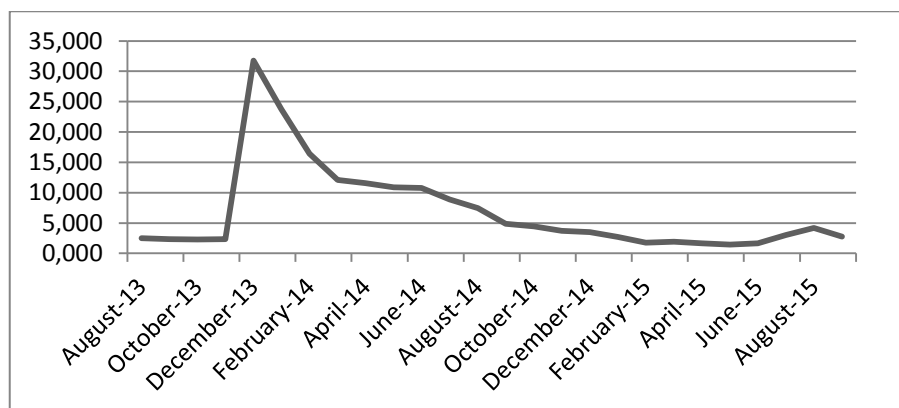


Figure 2 Litecoin exchange rate at BTC-e marketplace

Figure 2 shows Litecoin historical rates on BTC-e electronic marketplace in the same period of time as in the case of Bitcoin. Both charts have very similar shape indicating both cryptocurrencies having high volatility during given period of time.

In order to measure this volatility, available data on Bitcoin and Litecoin prices on electronic markets were gathered. Price changes between the two following trading days were calculated and percentage of change was expressed. As measure of volatility then was used standard deviation of all percentage price changes available. Annualized and thirty-day volatility were calculated by adjustment with square root rules. In order to compare volatility of these two major cryptocurrencies with volatility of official fiat currencies, two money pairs were selected.

Exchange rate of Euro and British pound to US dollar were used to have comparison on the same basis of price of currency in USD. Furthermore the comparison with gold price (in USD) changes was conducted to illustrate volatility to the precious that reflects non-fiat currency from the past. Following table shows us comparison of all calculated volatilities of cryptocurrencies, currency pairs and gold.

Table 1 Comparison of volatilities

Volatility Comparison	Daily Volatility	30-Day Volatility	Annualized Volatility
Bitcoin	9,10%	49,87%	144,53%
Litecoin	14,03%	77,88%	225,71%
Gold	0,91%	4,99%	14,47%
EUR/USD	0,45%	2,48%	7,18%
GBP/USD	1,97%	10,80%	31,30%

Source: Own processing

Our research showed that cryptocurrencies have extremely high volatility in comparison to major official currencies and gold. Volatilities of both investigated cryptocurrencies were incomparably higher than standard currencies or gold. High volatility therefore lowers trust in cryptocurrencies in comparison to official currency. On the other hand, even short-term speculations could bring high earnings to cryptocurrency speculators, what can paradoxically attract this type of investors. This led to big price growths and creation of price bubbles (e.g. in late 2013). Investors holding cryptocurrencies before volatile year of 2013 might have recorded above 1000% earnings when selling in at the peak of bubble. All these effects are transferred from Bitcoin and Litecoin also into other cryptocurrencies, because of their interchangeability to standard currencies only via medium of Bitcoin or Litecoin.

3.6 Security

Bitcoins are generated by mining procedures, where miners use special software to solve cryptographic hash problems and it is primary source to gain bitcoins. Miners are awarded with bitcoins only if they provide valid proof-of-work and this keeps BTC network stable, safe and secure by approving transactions. Proof-of-work is a process of verifying the block of data (transactions) in order to generate the next block by computing some random sequence (cryptographic problem), which is computed using a brute-force method employing software or hardware resource.

The first factor of cryptocurrency security is the security of block and block chains. A block of data with unverified cryptocurrency transactions has a unique key embedded. A new block cannot be submitted to the network without opening the present block of unverified transactions. When proof-of-work is found, the cryptocurrency network verifies whether the found key is correct. Every block contains a hash from its previous block and so blocks linked in this chronological order form the block. Each block in chain cannot be modified, because every block in chain after it would also have to be modified too. A valid block chain holds valid blocks and transactions only if it starts with the very first block.

Another aspect of cryptocurrencies' security is the security of generation algorithm. For example, creation of bitcoins is based on the SHA-256 hash algorithm to generate verifiably random sequences. It can be computed by using software or hardware resources and Elliptic Curve Digital Signature Algorithm, which is used for digital signing. A unique digital signature for each transaction allows a valid owner of bitcoin to spend funds from his account in secure way, while SHA-256 is used as block generation algorithm.

The third factor of security of cryptocurrencies is the network security. The possible risks of cryptocurrency networks were investigated by Reddy (2015) by network security analysis of Bitcoin. Malicious nodes can come into being with newly installed client, whose has no information of the block chain and it will always be vulnerable to attacks by connecting to attacker. Another type of attack is called packet sniffing. An attacker tracks all incoming and outgoing data during communication, then he can detect the transactions pertaining to any node and find addresses stored in the wallet and use them later to target specific machines with valuable wallets. Attacker might try to disturb the cryptocurrency protocol. Miners are encouraged to drop the connections with any node that is dealing with invalidated cryptocurrency data. This type of attack is known as Denial of service. Reddy (2015) identified also more possible disturbances of cryptocurrency networks, such as clock drifting, segmentation, dropping transactions, 51% attack, double spend attack etc.

Several initiatives are emerging to draft security framework for securing cryptocurrency systems and to address all aspects of cryptocurrency security. The standards that would help secure all information systems that make use of cryptocurrencies should be set. Draft of such standards was prepared by CryptoCurrency Certification Consortium (2014). If cryptocurrency-based businesses would retain these standards, using cryptocurrency might become more secure and trust in cryptocurrencies might increase.

4 Conclusions

Cryptocurrencies are very recent topic object in economy. Our research identified multiple factors affecting trust in cryptocurrencies. All of mentioned aspects affect trust in cryptocurrencies as long as their usage. The most of factors are interconnected others, have joint effects and together they affect trust in cryptocurrencies.

Enormous volatility of cryptocurrencies exchange rates was detected. Their high volatility causes high risk of trading cryptocurrency and is reflected in the formation of price bubbles. However, the great growths of their exchange rates attracted many speculators, but it is obvious that cryptocurrencies can only hardly retain their value. This fact can lead to change in understanding of cryptocurrencies as payment medium, but rather as specific commodities. Cryptocurrencies in comparison with commodities have advantage of easy portability thanks to their virtual character. However it makes them unusable outside of electronic environment. Cryptocurrencies are still popular medium of exchange in black economy. If potential legitimate users' trust in cryptocurrencies will rise in future, they will be used in increased scale also officially. Otherwise trust in cryptocurrencies might not reach necessary levels and their boom might sublime. Regulators around the world should create standards regarding cryptocurrency use that should be enforced by to increase security in the cryptocurrency industry and increase trust in cryptocurrencies.

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